

KaryoFlow: Few-Step Rectified-Flow Chromosome Detection With Localization-Quality Calibration

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Abstract—Automated karyotyping requires reliable detection of approximately 46 chromosomes from 24 morphologically similar classes in each metaphase image. Small objects, touching or overlapping instances, weak boundaries, and near-saturated class confidence make precise localization and reliable ranking particularly difficult. We present *KaryoFlow*, a detector designed for this regime through low-dimensional rectified box transport, identity-consistent refinement of adjacent proposals, and final localization-quality calibrated ranking. On two public chromosome datasets with markedly different object-scale distributions, *KaryoFlow* attains the highest observed mAP: it improves on the strongest evaluated detector by 2.2% on the small-object-heavy cohort and reaches a comparable point estimate (within 0.0001) on the scale-shifted cohort. Combining few-step inference with cascade-head distillation provides up to $2.73\times$ faster inference than the RF+Heun reference, while distillation reduces mAP by only 0.0003 relative to its six-head teacher. These results establish a localization-aware generative approach and provide a new direction for high-precision chromosome detection.

Index Terms—Rectified Flow, object detection, localization quality, diffusion models, medical image analysis, chromosome karyotyping

I. INTRODUCTION

CHROMOSOME karyotyping remains a labor-intensive step in cytogenetic analysis. Its detection stage is structurally unusual: a metaphase spread contains a nearly fixed set of about 46 chromosomes assigned to 24 classes, many instances occupy less than one percent of the image, and neighboring chromosomes frequently overlap in their bounding-box support. In our two held-out cohorts, the median instance count is 46; 99.1% and 100% of images contain at least one pair of ground-truth boxes with IoU above 0.1. The median relative box area also shifts from 0.4% to 0.8% between cohorts. Consequently, a few-pixel error can preserve AP50 while changing the strict-IoU ordering that determines mAP, and an iterative detector must refine many adjacent candidates without confusing their identities.

Conventional detectors offer complementary but incomplete solutions. Region-based Cascade R-CNN [1] and anchor-free YOLOX [2] provide strong conventional baselines, whereas transformer detectors such as DINO [3] obtain strong localization through multi-scale deformable attention [4]. DiffusionDet [5] instead treats boxes as a set generated from

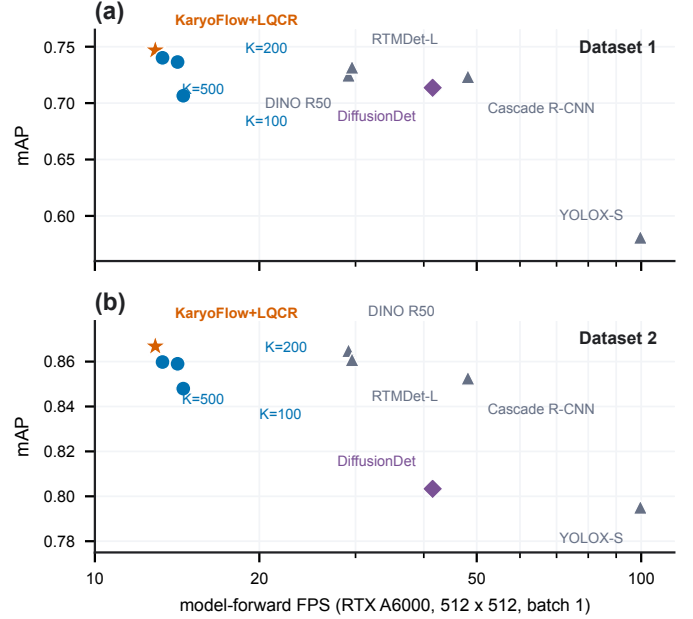


Fig. 1. Speed-accuracy comparison on two chromosome cohorts. Accuracy follows the prespecified evaluation protocol; model-forward FPS is measured with the same 512×512 , batch-1 RTX A6000 protocol. Accuracy points in both panels are three-inference-seed means from fixed trained models. Both panels include three unconnected RF-DPM++ operating points at $K = 500$, 200, and 100; the Dataset 1 $K = 200/100$ points use inference-only pruning of the same model. Higher and farther right is better.

noise, an appealing formulation for variable spatial layouts, but its DDPM inheritance introduces a curved stochastic path [6], while additional refinement steps increase the cost of time-sensitive microscopy analysis. More importantly, simply shortening that path does not resolve two detection-specific mismatches: multistep integration assumes a persistent state identity, whereas proposal renewal replaces selected states; and final ranking by class confidence ignores whether a small chromosome is localized accurately enough to survive strict IoU thresholds.

Rectified Flow [7] (RF) offers a natural generation foundation by replacing the DDPM path with a deterministic linear transport between noise boxes and targets. Applying RF to

detection raises three linked questions: how to learn a well-conditioned box trajectory from limited data, how to integrate it without violating proposal identity, and how to rank fixed outputs by both semantic and geometric reliability. *KaryoFlow* organizes the answer as Generation, Decision, and

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Deployment, shown in Figure 2. Its contributions are:

First, we adapt few-step box transport to high-instance-count metaphase detection. The generation stage combines RF in normalized four-dimensional box space, a direct data-prediction objective, and identity-consistent multistep integration. We show that replacing a proposal invalidates the history secant unless its history is reset, providing a design principle for crowded box-set refinement. RF and DPM-Solver++ [8] are adopted foundations; the contribution is their task-specific formulation and analysis.

Second, we introduce Localization-Quality Calibrated Ranking (LQCR) for the strict-IoU bottleneck encountered after class confidence has nearly saturated. LQCR learns localization reliability from final proposal features and ranks detections by $s = pq^2$. It is applied only when outputs are emitted, leaving candidate coordinates, class scores, proposal survival, and the RF trajectory unchanged before final post-processing. Here, calibration denotes calibration of the ranking score to localization quality, not probabilistic calibration.

Third, the deployment layer separates solver evaluations from cascade-stage evaluations and compresses them through few-step integration, head distillation, and proposal pruning. We evaluate all three layers on two public chromosome datasets containing 2,200 and 5,000 images using independent training runs, paired component comparisons, held-out tests, strict-IoU analysis, and standardized latency profiling.

II. RELATED WORK

a) Chromosome detection and karyotyping: A metaphase image combines several difficulties rarely present together in natural-image detection: dozens of adjacent objects, homologous classes with similar morphology, scale variation between chromosome groups, staining shift, and incomplete boundaries caused by touching or overlap. Earlier work addressed joint segmentation–classification [9], single-chromosome classification [10], chromosome enumeration [11], straightening [12], and detection from metaphase images [13]. Recent 2026 studies confirm that the clinical workflow is moving toward direct detection from whole metaphase images: object detectors remove manual single-chromosome cropping [14], and integrated pipelines combine YOLO-based detection with pairing and anomaly screening [15]. However, those reports emphasize concordance, mAP50, or downstream workflow accuracy. These measures can remain high when a small chromosome has a several-pixel boundary error. A two-step transfer study across Q- and G-band chromosome datasets further shows that gains depend on staining-domain similarity and target-data quality [16], while an integrated karyotyping pipeline explicitly excludes morphologically unreliable, overlapped metaphases before detection [15]. KaryoFlow therefore targets the localization stage itself, with mAP over IoU 0.50:0.95, strict-IoU diagnostics, and measured inference cost.

Large-scale representation learning addresses a complementary limitation. CHROMA was pretrained on more than four million chromosome images and improves robustness to rare abnormalities and challenging acquisition conditions [17]. Such single-chromosome representations reduce annotation burden after extraction, but do not resolve how densely

overlapping candidates should move, retain identity, or be ranked in the preceding box-detection stage. This distinction motivates a detector specialized for metaphase box geometry rather than another isolated-chromosome classifier.

b) Generative detection and few-step integration: DiffusionDet [5] formulates object detection as iterative box denoising. FlowDet [6] and DeFloMat [18] replace DDPM-style dynamics with conditional flow matching or RF, while DiffuBox [19] studies a distinct 3D refinement setting. RF [7] and Flow Matching [20] provide our adopted continuous-time foundation, and DPM-Solver++ [8] supplies the multistep integration principle. Recent work has also explored flow-matching design and optimized priors for cellular-microscopy generation and medical augmentation [21], [22]; these image-generative settings are complementary to our discriminative transport of box sets. Our focus is complementary: KaryoFlow derives the data-prediction update for the RF box path and makes proposal identity an explicit condition of multistep inference. Representative recent TMI diffusion studies address synthesis, segmentation, or robust classification [23]–[25], leaving few-step medical box-set prediction comparatively underexplored.

c) Localization quality in medical detection: Medical detectors must handle small targets, incomplete labels, acquisition shift, and structured false positives. Representative TMI studies address few-shot pulmonary lesions, heterogeneous lesion annotations, and relational transfer across domains [26]–[28], while context refinement, weak supervision, and imbalanced learning reduce task-specific false positives [29]–[31]. For general detection, IoU-Net predicts localization confidence for proposal selection [32], and GFL and VarifocalNet combine class and localization quality in dense heads [33], [34]. LQCR inherits the probability-ranking motivation but differs in scope: it is applied only after a generative detector has completed all trajectory updates, so solver history, coordinates, predicted classes, and proposal survival are fixed. This controlled boundary is particularly useful in chromosome images, where AP50 is nearly saturated but strict-IoU errors remain. Its ranking role is distinct from probabilistic calibration, which has been studied separately in medical image segmentation [35].

III. METHOD

KaryoFlow separates three roles that are easy to conflate in iterative detection. The *Generation* layer transports a persistent set of noisy boxes to chromosome boxes. The *Decision* layer ranks the completed candidates by semantic and localization reliability. The *Deployment* layer reduces solver and cascade-head computation without changing either role.

A. Generation: Few-Step Identity-Consistent Box Transport

1) Box-Space Rectified Flow:

a) Linear transport in chromosome-box space: Each chromosome is represented by a normalized center and size vector (c_x, c_y, w, h) . Training connects a Gaussian source box $\mathbf{x}_1$ to a matched chromosome box $\mathbf{x}_0$ through

$$\mathbf{x}_t = (1 - t) \mathbf{x}_0 + t \mathbf{x}_1, \quad t \in [0, 1], \quad (1)$$

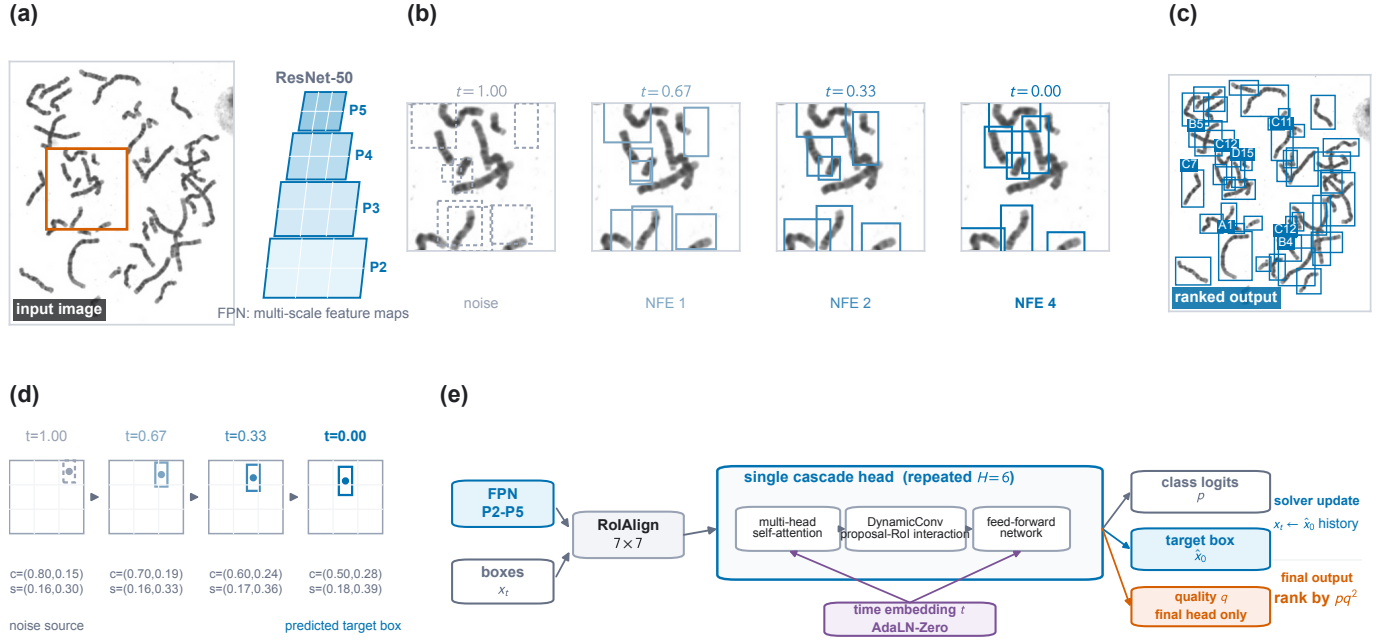


Fig. 2. KaryoFlow from image to detections. (a) A metaphase image and its ResNet-50-FPN representation provide multi-scale image features. (b) A dense crop visualizes four-NFE box refinement from noise to localized chromosome candidates. (c) The detector emits the final quality-ranked chromosome detections. (d) One proposal is shown in normalized center-and-scale coordinates along the prescribed linear RF path. (e) At each evaluation, multi-scale RoIAlign feeds a cascade head with self-attention, DynamicConv, feed-forward refinement, AdaLN-Zero time conditioning, and the final-head quality branch. Endpoints and output boxes are empirical; intermediate boxes visualize the prescribed RF path rather than recorded hidden activations.

whose target velocity is $\mathbf{x}_1 - \mathbf{x}_0$. Unlike image diffusion, only four box coordinates are transported; the microscopy features remain fixed conditioning information. This low-dimensional, straight path is a natural match for a metaphase spread, where approximately 46 boxes must move concurrently while preserving their individual identities.

b) Time conditioning: AdaLN-Zero [36] injects the time embedding into each proposal head through zero-initialized residual modulation. It is retained as a stable conditioning mechanism; controlled ablation does not support treating it as a separate accuracy contribution.

2) Stable Parameterization and Few-Step Integration:

a) Few-step integration: Euler integration provides the first-order reference: it is inexpensive per step but uses only the current box estimate. Heun adds a predictor-corrector evaluation to reduce local integration error, so four reverse steps require seven network evaluations. DPM-Solver++ [8] instead reaches second order by reusing the previous target-box prediction, requiring only one new evaluation per step. We compare the solvers in this progression—Euler, Heun, then DPM-Solver++—and attribute solver gains only under matched trained models and step counts.

The model directly predicts $\mathbf{x}_0$ (the target box), and the update uses polynomial interpolation of the $\mathbf{x}_0$ history in t space (not the log-SNR λ space of VP-SDE). The RF-ODE in data-prediction form is $d\mathbf{x}/dt = (\mathbf{x} - \mathbf{x}_0(t))/t$; integrating exactly over $[t_n, t_{n+1}]$ with linear $\mathbf{x}_0(t)$ interpolation yields the update in (2)

$$\mathbf{x}_{t_{n+1}} = \frac{t_{n+1}}{t_n} \mathbf{x}_{t_n} + \left(1 - \frac{t_{n+1}}{t_n}\right) \mathbf{x}_0^{(n)} + \varphi_1 \mathbf{D}_1, \quad (2)$$

$$\varphi_1 = t_{n+1} \log \frac{t_n}{t_{n+1} - t_n + t_{n+1}}, \quad \mathbf{D}_1 = \frac{\mathbf{x}_0^{(n)} - \mathbf{x}_0^{(n-1)}}{t_n - t_{n-1}},$$

The first two terms move a proposal toward its current box estimate; the last term corrects that move using the change between the two most recent estimates. The initial noise sample remains fixed. Thus four steps require four rather than seven network evaluations. The endpoint is protected with $t_{n+1} > 10^{-7}$.

b) Why direct data prediction is numerically preferable here: For the path in (1), $\mathbf{x}_0 = \mathbf{x}_t - t\mathbf{v}$. Hence the two parameterizations carry the same information, but their pointwise squared errors satisfy

$$\mathcal{L}_v(t) = t^{-2} \mathcal{L}_{x_0}(t). \quad (3)$$

With the shifted schedule, velocity regression therefore amplifies gradient noise as $t \rightarrow 0$, where accurate terminal boxes matter most. Direct x_0 prediction removes this avoidable conditioning factor. We use this parameterization for its better-conditioned terminal objective, without claiming a test-set accuracy gain over algebraically equivalent velocity prediction.

3) History-Consistent Proposal Integration:

a) Proposal identity is a premise of multistep history: Let $r_i^n = 1$ indicate that proposal row i is renewed after step n .

Proposition 1 (Identity consistency). *The multistep correction $\mathbf{D}_{1,i}$ in (2) is a consistent first-order approximation to a trajectory derivative only if the two states indexed by row i belong to the same proposal trajectory, or if the row-wise history is reset after replacement.*

Proof. Without renewal, differentiability of $\mathbf{x}_{0,i}(t)$ gives the

standard secant limit

$$\mathbf{D}_{1,i} = \frac{\mathbf{x}_{0,i}^{(n)} - \mathbf{x}_{0,i}^{(n-1)}}{t_n - t_{n-1}} \rightarrow \frac{d\mathbf{x}_{0,i}}{dt}. \quad (4)$$

If $r_i^n = 1$, the numerator subtracts predictions from two independently initialized states. Its limit contains a finite cross-trajectory jump divided by $t_n - t_{n-1}$ and therefore is not the derivative along either trajectory. Resetting history removes the invalid correction. $\square$

We use the simpler alternative of disabling renewal after verifying sufficient proposal redundancy. This choice is well matched to metaphase images: the object count is tightly concentrated near 46, so $K = 500$ provides multiple candidates per chromosome, while maintaining identity avoids mixing adjacent, morphologically similar instances. In the prespecified Dataset 2 evaluation, disabling renewal at $K = 100$ costs 0.0169 mAP relative to retaining renewal, showing that history consistency cannot compensate for inadequate capacity.

B. Decision: Localization-Quality-Aware Ranking

Chromosome classes can be predicted confidently even when their boxes are a few pixels too loose or shifted. For small chromosomes, a few-pixel displacement produces a large IoU change; for touching chromosomes, neighboring candidates can have similar class confidence despite different localization quality. For proposal feature F , correct-class event C , IoU U , and evaluation threshold τ , the true-positive probability separates as

$$P(C=1, U \geq \tau | F) = P(C=1 | F)P(U \geq \tau | C=1, F). \quad (5)$$

The second factor motivates augmenting class confidence with localization information. LQCR does not estimate this threshold event probability directly; instead, it learns continuous IoU as a ranking surrogate through a two-layer quality branch $q = \sigma(g_\phi(F))$ attached only to the final cascade stage. It adds 33,153 parameters. The backbone, FPN, proposal interaction, class head, and box head are frozen during branch fitting, so the experiment isolates the ranking-stage effect from box generation. Here calibration means alignment of the detection ranking with localization quality; it is not presented as calibrated posterior probability.

For Hungarian-matched proposal i , let $m_i = 1$ and let u_i be the detached aligned IoU; unmatched proposals have $m_i = 0$. We train the scalar logit with the continuous target $y_i = m_i u_i$ and weighted binary cross-entropy

$$\mathcal{L}_q = \frac{\lambda_q}{\max(N_+, 1)} \sum_i w_i \ell_{\text{BCE}}(g_i, y_i), \quad (6)$$

$$w_i = m_i u_i + (1 - m_i) \alpha q_i^\gamma,$$

where $\lambda_q = 0.25$, $\alpha = 0.75$, and $\gamma = 2$. Because the detector weights are frozen and the IoU targets are detached, fitting the quality branch cannot update predicted box geometry. Matched proposals are supervised toward their aligned IoU, whereas unmatched proposals receive a zero target. At output, class probability p is replaced by $s = pq^2$. The exponent is an empirically selected ranking transform, not a consequence

of (5). A validation sweep over $\beta \in \{0, 0.25, 0.5, 1, 2\}$ selects the exponent 2; this learned-quality sweep is separate from the true-IoU upper bound used only to quantify headroom.

The final-only design keeps raw class scores in the solver, proposal renewal, Top- K pruning, and early stopping. Calibration is applied only when emitting detections, so boxes, predicted classes, and RF trajectories remain fixed at the candidate level; score-dependent NMS or max-detection selection may consequently choose a different final subset. All detector parameters are frozen while the quality branch is fitted, isolating ranking from box generation. This design targets the observed chromosome regime in which AP50 is nearly saturated while substantial error remains at strict IoU thresholds. Figure 3 summarizes the ranking principle, its final-stage operating scope, and the paired fixed-model effect.

C. Deployment: Orthogonal Inference Compression

a) *Cascade-head distillation*: Solver NFE counts network calls, whereas each call evaluates H cascade heads. We distill a six-stage teacher into a three-stage student with stage-aligned feature and prediction supervision. At four solver steps this reduces cascade-head evaluations (CHE) from 24 to 12, while NFE remains four; the reduction is therefore 24-to-12 CHE, not NFE. The compressed model reaches 0.8595 versus 0.8598 and reduces RTX A6000 latency from 75.20 to 44.10 ms. Together with few-step integration, this reduces latency from the 120.34-ms RF+Heun reference to 44.10 ms, a $2.73\times$ acceleration; the 0.0003 mAP cost is measured against the six-stage teacher under the same three-inference-seed test protocol.

b) *Top- K proposal pruning*: After step 0 of inference, we prune proposals from 500 to K based on confidence scores; only the top- K proposals proceed through steps 1–3. Because pruning changes the proposal set, multistep history is reset before the retained proposals continue refinement.

IV. EXPERIMENTS

A. Experimental Setup

a) *Datasets*: Table I summarizes the two public chromosome datasets, referred to as Dataset 1 and Dataset 2. Dataset 1 is Chromosome20240904 (RST) [37], with 2,200 images; Dataset 2 is the 24 Chromosomes Object dataset [38], [39], with 5,000 images. Their training splits contain 1,540 and 3,500 images, respectively. Both have a median of 46 annotated instances per image, but their median relative box areas differ substantially (0.4% versus 0.8%). Under the COCO absolute-area definition, 23.0% of Dataset 1 instances are small, compared with only 0.4% in Dataset 2; correspondingly, 100% and 11.3% of test images contain at least one small instance. Moreover, 99.1% and 100% of test images contain at least one ground-truth box pair with IoU above 0.1. These statistics quantify the small, crowded regime and the scale shift used for cross-dataset replication. Both datasets use image-level random splitting. A single blood sample may yield multiple metaphase images, but patient identifiers are unavailable; thus patient-level separation cannot be verified. We analyze only the publicly released images and annotations and collect no additional human-subject data.

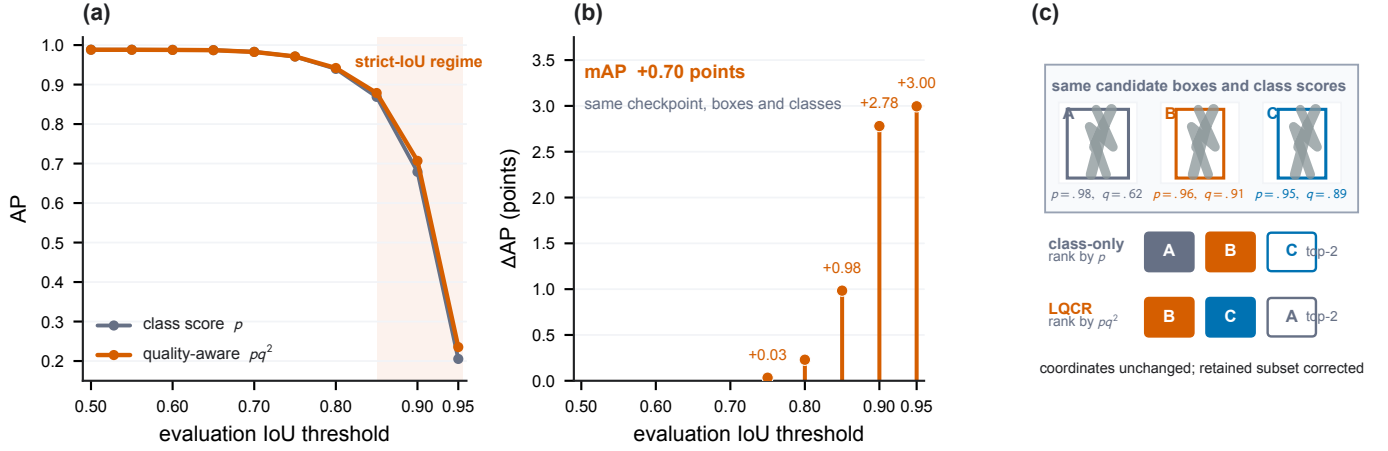


Fig. 3. Effect and operating scope of localization-quality ranking. (a) On the same Dataset 2 RF-DPM++ model, quality-aware and class-only AP curves separate mainly at strict IoU thresholds. (b) Their threshold-wise difference reaches +2.78/+3.00 points at AP90/AP95, while AP50 changes by only -0.01 points. (c) The same candidate boxes and class scores are shown before class-only and LQCR ranking; localization quality promotes the tighter candidate and changes the retained top-2 subset without altering candidate geometry. Panels (a)–(b) report held-out measurements; (c) illustrates the ranking mechanism.

TABLE I
DATASETS USED IN THIS PAPER. DATASET 1 IS CHROMOSOME20240904 (RST); DATASET 2 IS THE 24 CHROMOSOMES OBJECT DATASET.

Dataset	Train	Val	Test	Classes
Dataset 1	1,540	440	220	24
Dataset 2	3,500	500	1,000	24

TABLE II
HELD-OUT-TEST PROPERTIES THAT MOTIVATE THE DETECTOR DESIGN. RELATIVE AREA IS BOX AREA DIVIDED BY IMAGE AREA; SMALL FOLLOWS THE COCO ABSOLUTE-AREA THRESHOLD ($< 32^2$ PIXELS); OVERLAP DENOTES AN IMAGE CONTAINING AT LEAST ONE PAIR OF GROUND-TRUTH BOXES WITH $\text{IoU} > 0.1$.

Dataset	Median n	Med. rel. area	Small inst.	Images w/ small	Overlap
Dataset 1	46	0.4%	23.1%	100%	99.1%
Dataset 2	46	0.8%	0.4%	11.3%	100%

b) Architecture and training: Our base framework (KaryoFlow) uses ResNet-50 [40] + FPN [41] (256 channels, 4 levels), 500 proposals, 6 cascade transformer heads with deep supervision (5 auxiliary heads). Optimization: AdamW [42] (learning rate 5×10^{-5} , weight decay 10^{-4}), 5-epoch warmup + CosineAnnealing [43], 150 epochs. Loss: Focal [44] ($\lambda_{\text{cls}}=2$) + L1 ($\lambda=5$) + GIoU [45] ($\lambda=2$) with Hungarian matching [46]. Diffusion uses RF with shifted noise schedule (shift=3); default solver is DPM-Solver++ 4-step. For the isolated LQCR intervention, the trained six-stage detector initializes the model, all pre-existing parameters are frozen, and only the 33k-parameter quality branch is optimized for at most 12 epochs with AdamW learning rate 10^{-3} and cosine decay; model selection uses validation performance. Thus this phase fits a ranking statistic but cannot alter the detector’s generated boxes or class logits.

c) Detection protocol: Boxes are represented in two spaces: image space (absolute pixel $xyxy$) for RoIAlign and NMS, and diffusion space ($cxcywh$ normalized to $[0, 1]$, mapped to $[-s, +s]$, $s=2$). The RF forward path is $x_t =$

$(1-t)x_0 + t\varepsilon$ (DDPM baseline uses cosine schedule). At inference, 500 random-noise proposals are iteratively denoised; with time-ensemble, predictions from all steps are concatenated and deduplicated by per-class NMS (IoU 0.5). Evaluation uses COCO mAP@0.50:0.95 (10 IoU thresholds, maxDets=100). Validation data are used only for model and hyperparameter selection. Unless explicitly identified as a diagnostic upper bound, every final detector comparison, post-selection hyperparameter sensitivity analysis, and accuracy coordinate in an efficiency plot is evaluated on the held-out test cohort (220 images for Dataset 1 and 1,000 for Dataset 2). Accuracy evaluation is hardware independent under the fixed software and seed protocol; all latency and FPS measurements use the same RTX A6000.

d) Statistical considerations: Three independently trained detectors with distinct recorded framework seeds are evaluated on each dataset. LQCR is fitted separately to each frozen parent detector and evaluated with the same inference seed within every pair. Fixed-checkpoint inference-seed means remain only in mechanism and efficiency analyses, where they quantify proposal-initialization variation rather than training uncertainty.

e) Reproducibility: All reported comparisons retain their dataset split, random seed, trained model, and evaluation protocol. Controlled results are separated from exploratory runs and diagnostic upper bounds. For LQCR, the base detector is frozen and learned-quality sweeps are distinguished from analyses using true IoU. This protocol prevents unmatched training conditions or unattainable upper bounds from entering model claims.

B. Main Results: RF vs DDPM

a) Dataset 2 ablation: Table III reports two complementary Dataset 2 comparisons. The first pair measures the change from DDPM to the RF generation foundation under the corresponding trained models. The second pair uses one trained RF-DPM++ model to isolate the final-only LQCR intervention.

Solver efficiency is evaluated separately in Table VIII using the same trained RF model across solvers. AdaLN-Zero is used throughout as time conditioning but is not counted as an independent contribution.

RF supplies the dominant historical improvement over the DDPM baseline. DPM-Solver++ retains accuracy while reducing Heun’s seven evaluations to four; paired solver tests do not support an additional precision claim. LQCR then improves mAP by 0.0070 in the paired three-inference-seed test without changing the detector, candidate coordinates, class logits, renewal decisions, or trajectory. Its largest gains occur at strict IoU thresholds (AP90/AP95 +0.0278/ +0.0300), consistent with the localization-aware ranking mechanism rather than a coordinate change.

Head distillation is reported separately in the deployment benchmark because the three-head student was trained only on Dataset 2. Validation-only and single-seed exploratory directions are excluded from this table.

b) Dataset 1: RF vs DDPM: On the Dataset 1 held-out test split (3 seeds), RF+Heun reaches 0.7367 ± 0.0015 mAP versus 0.7187 ± 0.0031 for DDPM, an absolute gain of 0.0180. On Dataset 2, the unified held-out test comparison gives 0.8566 for RF+Heun and 0.8033 for DDPM, an absolute gain of 0.0533. These comparisons use matched dataset splits and evaluation protocols.

C. SOTA Comparison

Table V uses three independently trained KaryoFlow detectors on each chromosome dataset. The cohorts differ strongly in relative object scale. KaryoFlow+LQCR obtains 0.7480 mAP on Dataset 1 and 0.8650 on Dataset 2; the corresponding margins over the strongest listed alternatives are 0.0161 and 0.0001. The latter is a highest point estimate rather than evidence of a material advantage over DINO. The result is not caused by a larger backbone: KaryoFlow uses ResNet-50-FPN, whereas RTMDet-L uses CSPNeXt-L and DINO uses multi-scale deformable attention [3], [47]. We therefore describe these as best results within the evaluated comparison, rather than claiming an exhaustive cross-study SOTA.

Because the Dataset 2 margin is only 0.0001, we additionally performed a paired image-cluster bootstrap on the common 1,000-image test cohort. Across 1,000 deterministic replicates, the 95% percentile interval for the LQCR–DINO mAP difference is $[-0.0033, 0.0022]$. This interval quantifies image-sampling uncertainty conditional on the evaluated checkpoints; DINO has only fixed-model inference replications, so it does not measure between-training uncertainty. The result supports statistical comparability, not a Dataset 2 superiority claim.

a) Small-object performance: Dataset 1 is the small-object-heavy cohort. Across three independently trained detector pairs, the LQCR mean is 0.7480 ± 0.0017 mAP and 0.5147 ± 0.0090 AP_S, improving the uncalibrated KaryoFlow means by 0.0067 and 0.0057 and exceeding the strongest listed AP_S comparator by 0.0142. On Dataset 2, LQCR yields the highest overall mAP point estimate but remains below DINO on AP_S. Across the three paired Dataset 2 runs, it

changes AP_S by -0.0020 ± 0.0029 while improving mAP by 0.0065 ± 0.0004 . This pattern supports the intended role of quality calibration—better aggregate ordering across strict IoU thresholds—without claiming universal superiority at every scale.

b) Class- and scale-resolved analysis: The held-out tests show that AP50 is already near saturation, whereas the quality-ranking gain concentrates at AP90/AP95. The remaining headroom therefore lies in precise coordinate refinement and localization-aware ordering rather than simply producing more low-quality proposals. This pattern is especially relevant to the smaller object distribution in Dataset 1.

c) Interpretation of paired comparisons: The paired solver comparison is accuracy-neutral. The retained positive intervention is LQCR: enabling its score only at the final output changes the three-inference-seed mean mAP from 0.8598 to 0.8668 on exactly the same trained detector and candidate detections before ranking. This construction isolates the ranking-stage effect from generation, interaction, and numerical integration effects. We report this as a controlled paired intervention, not as a training-seed replication; the separate SOTA comparison supplies independent-training evidence on both datasets.

The monotone test-set trend is consistent with the validation-selected $\beta = 2$ and the intended strict-IoU ranking behavior.

1) Scale-Resolved Cross-Dataset Comparison: Figure 4 replaces selected visual cases with complete-set evidence. Dataset 1 has a substantially smaller relative-box distribution than Dataset 2 (median 0.4% versus 0.8%). On the Dataset 1 test split, KaryoFlow+LQCR reaches AP_S = 0.5147 ± 0.0090 over three independently trained detector pairs, a 0.0057 gain over the uncalibrated model and 0.0142 above the strongest listed comparator. On Dataset 2, the three-pair means are AP_S = 0.5801 with LQCR and 0.5821 without it; both remain below DINO R50 (0.5953). The result supports scale-dependent small-object competitiveness rather than a universal advantage. Because COCO’s small-area threshold is absolute, raw AP_S levels are interpreted only within each dataset; the relative-area CDF is used to characterize the cross-dataset scale shift.

2) Difficulty-Stratified Decision Analysis: To test whether the strict-IoU gain is concentrated in difficult instances, we performed a post-hoc paired analysis of the three Dataset 2 detector pairs. Ground-truth instances were stratified by relative-area quartile and by the maximum IoU with another ground-truth chromosome. Table VII reports class-consistent, one-to-one recall changes at a fixed top-100 operating point. This metric avoids the false-positive artifacts that would arise from computing AP after filtering the ground-truth set.

The largest strict-threshold response occurs in Q2 at R_{95} , and the moderate- and high-overlap strata both improve at R_{90} and R_{95} . The standard COCO-small subset contains only 166 Dataset 2 instances and gives an unstable $\Delta R_{90} = -0.0080 \pm 0.0092$; we therefore do not claim a universal small-object advantage. The supported interpretation is narrower: LQCR improves strict-IoU ranking overall, with descriptive gains in lower-area quartiles and in neighboring-box overlap strata.

TABLE III

DATA SET 2 COMPONENT ANALYSIS. THE FIRST PAIR COMPARES THE DDPM AND RF GENERATION FOUNDATIONS. THE FINAL PAIR USES THE SAME TRAINED RF-DPM++ MODEL AND REPORTS THREE-INFERENCE-SEED TEST MEANS; LQCR CHANGES ONLY OUTPUT RANKING. THE TWO PANELS ANSWER DIFFERENT QUESTIONS AND DO NOT CONSTITUTE A CUMULATIVE ABLATION. SOLVER EFFECTS ARE REPORTED SEPARATELY UNDER A PAIRED PROTOCOL.

Experiment	Solver	Steps	NFE	mAP	AP50	AP75	AP _S	AP _M	AP _L
<i>Generation foundation (separately trained models)</i>									
DDPM baseline (Euler 1-step)	Euler	1	1	0.8033	0.9714	0.9381	0.5323	0.8006	0.8221
RF+Heun (KaryoFlow)	Heun	4	7	0.8566	0.9878	0.9691	0.5827	0.8541	0.8496
<i>Final-stage decision (one fixed detector)</i>									
DPM-Solver++	DPM++	4	4	0.8598	0.9883	0.9709	0.5676	0.8572	0.8876
+LQCR (final-only)	DPM++	4	4	0.8668	0.9882	0.9714	0.5663	0.8640	0.8890

TABLE IV

CORE INTERVENTIONS ON BOTH HELD-OUT TEST COHORTS. LQCR VALUES ARE PAIRED MEANS OVER THREE INDEPENDENTLY TRAINED PARENT DETECTORS PER DATASET. RENEWAL AND LATENCY VALUES USE THE FIXED-DETECTOR DEPLOYMENT PROTOCOL.

Intervention	D1 Δ mAP	D2 Δ mAP	Deployment effect	Role and boundary
LQCR	+0.0067	+0.0065	+3.3% latency	Final-only ranking; candidate boxes and trajectories are unchanged
Renewal off ($K = 500$)	-0.0016	-0.0004	-2.4% D2 latency	Preserves DPM++ history identity; not claimed as an accuracy gain

TABLE V

HELD-OUT-TEST COMPARISON ON TWO CHROMOSOME DATASETS. KARYOFlow AND KARYOFlow+LQCR ARE MEANS OVER THREE INDEPENDENTLY TRAINED DETECTORS ON EACH DATASET; EACH LQCR MODEL IS PAIRED WITH ITS FROZEN PARENT. OTHER DETECTORS ARE FIXED-MODEL POINT ESTIMATES UNDER THE SAME SPLIT AND EVALUATOR. BOLD DENOTES THE BEST POINT ESTIMATE PER METRIC WITHIN THIS COMPARISON.

Method	Backbone	mAP	AP50	AP75	AP _S	AP _M	AP _L
<i>Dataset 1</i>							
KaryoFlow+LQCR	ResNet-50	0.7480	0.9323	0.8330	0.5147	0.7287	0.6540
KaryoFlow	ResNet-50	0.7413	0.9323	0.8267	0.5090	0.7240	0.6533
DINO R50	ResNet-50	0.7248	0.9341	0.8135	0.4853	0.7112	0.6244
RTMDet-L	CSPNeXt-L	0.7319	0.9344	0.8387	0.4803	0.7253	0.6221
Cascade R-CNN	ResNet-50	0.7235	0.9221	0.8307	0.5005	0.7088	0.6488
DiffusionDet	ResNet-50	0.7136	0.9078	0.7975	0.4618	0.7041	0.6230
YOLOX-S	CSPDarkNet-S	0.5811	0.9177	0.6920	0.3298	0.5777	0.5375
<i>Dataset 2</i>							
KaryoFlow+LQCR	ResNet-50	0.8650	0.9880	0.9693	0.5801	0.8623	0.8873
KaryoFlow	ResNet-50	0.8585	0.9880	0.9687	0.5821	0.8560	0.8817
DINO R50	ResNet-50	0.8649	0.9890	0.9728	0.5953	0.8617	0.9196
RTMDet-L	CSPNeXt-L	0.8609	0.9890	0.9706	0.5725	0.8584	0.8713
Cascade R-CNN	ResNet-50	0.8527	0.9850	0.9670	0.5724	0.8499	0.8745
DiffusionDet	ResNet-50	0.8033	0.9714	0.9381	0.5323	0.8006	0.8221
YOLOX-S	CSPDarkNet-S	0.7951	0.9854	0.9427	0.4861	0.7928	0.7895

TABLE VI

POST-SELECTION SENSITIVITY ANALYSIS OF THE LEARNED-QUALITY EXPONENT ON DATASET 2. VALUES ARE THREE-INFERENCE-SEED MEANS FOR ONE FIXED DETECTOR. VALIDATION SELECTED $\beta = 2$; TEST RESULTS ARE REPORTED TRANSPARENTLY AND WERE NOT USED FOR SELECTION. $\beta = 0$ DISABLES QUALITY RANKING.

β	mAP	Δ mAP	AP90	AP95
0	0.8598	—	0.6791	0.2053
0.25	0.8622	+0.0024	0.6890	0.2153
0.5	0.8638	+0.0040	0.6946	0.2221
1	0.8654	+0.0056	0.7014	0.2290
2	0.8668	+0.0070	0.7069	0.2353

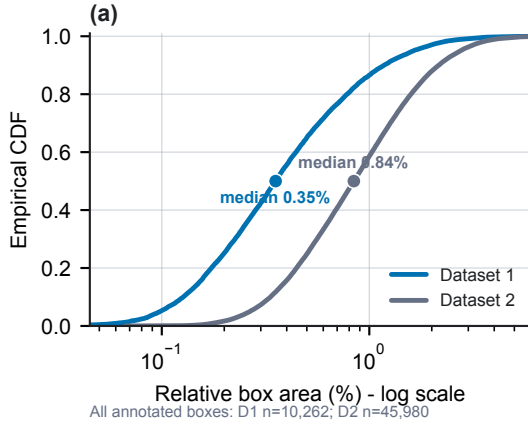
TABLE VII

POST-HOC DATASET 2 DIFFICULTY ANALYSIS OVER THREE INDEPENDENTLY TRAINED DETECTOR PAIRS. VALUES ARE PAIRED MEAN CHANGES $\pm$ SAMPLE SD. Q1/Q2 DENOTE THE TWO LOWER RELATIVE-AREA QUANTILES; O0, O2, AND O3 DENOTE MAXIMUM NEIGHBORING-GT IOU RANGES $[0, 0.01]$, $[0.10, 0.30]$, AND $[0.30, 1]$, RESPECTIVELY.

Stratum	Instances	ΔR_{90}	ΔR_{95}
Q1 (smallest 25%)	11,495	+0.0015 $\pm$ 0.0013	+0.0015 $\pm$ 0.0020
Q2 (25-50%)	11,495	+0.0010 $\pm$ 0.0005	+0.0046 $\pm$ 0.0006
O0 (non-overlap)	5,068	+0.0006 $\pm$ 0.0021	+0.0024 $\pm$ 0.0015
O2 (moderate)	21,835	+0.0006 $\pm$ 0.0006	+0.0031 $\pm$ 0.0014
O3 (high)	6,424	+0.0018 $\pm$ 0.0016	+0.0020 $\pm$ 0.0016

D. Solver Analysis

a) *DPM-Solver++ step ablation*: DPM-Solver++ improves from 0.8514 at one step to 0.8568 at four steps, consistent with rapid convergence along the short RF box trajectories.



(b)

Measure	Dataset 1	Dataset 2
Small-instance share	23.0%	0.4%
KaryoFlow AP_S	0.5090	0.5821
KaryoFlow+LQCR AP_S	0.5147 $\pm$ 0.0090	0.5801 $\pm$ 0.0123
LQCR ΔAP_S	+0.0057	-0.0020
Best comparator	Cascade R-CNN	DINO R50
Gap to comparator in AP_S	+0.0142	-0.0152
Overall mAP rank	1st	1st

AP_S is interpreted within each cohort; both are 3-training-run means.

Fig. 4. Object-scale shift and small-object detection across two chromosome cohorts. (a) The empirical CDF of relative ground-truth box area over all annotations shows that Dataset 1 chromosomes are markedly smaller than those in Dataset 2. (b) A compact scale-response table contrasts the small-instance prevalence, LQCR effect, strongest comparator, and overall mAP rank. AP_S is interpreted within, not across, datasets; both KaryoFlow columns are means over three independently trained detector pairs.

TABLE VIII

SOLVER $\times$ STEP COMPARISON USING ONE TRAINED RF MODEL. VALUES ARE THREE-INFERENCE-SEED MEANS ON THE DATASET 2 HELD-OUT TEST; MATCHED-STEP SOLVER DIFFERENCES ARE NEGLIGIBLE.

Solver	Steps	NFE	mAP
Euler	1	1	0.8513
Euler	4	4	0.8567
Heun	4	7	0.8566
DPM-Solver++	1	1	0.8514
DPM-Solver++	4	4	0.8568

b) *DPM-Solver++ vs Heun at matched NFE*: At four matched steps on the same trained model, DPM-Solver++ gives 0.8568, Heun 0.8566, and Euler 0.8567. The largest solver gap is only 0.0002, well below the observed inference-seed variation. We therefore attribute no accuracy gain to the off-the-shelf solver; its value is reducing the four-step Heun computation from seven to four network evaluations.

E. FPS / Latency Benchmark

Table IX and Figure 1 report the speed-accuracy trade-off at 512×512 input resolution. Latency is CUDA-event model-forward time and excludes image decoding, host preprocessing, and visualization. The unpruned $K = 500$ point and the $K = 200/100$ pruning points share one trained model and evaluation protocol. DPM-Solver++ with Top- K pruning reaches a latency compatible with interactive use, while single-shot detectors (Cascade R-CNN, YOLOX-S) trade accuracy for speed.

DPM-Solver++ +Top- K ($K=200$) reaches 70.57 ms / 14.2 FPS at mAP 0.8590. The complete KaryoFlow+LQCR model reaches 77.68 ms / 12.9 FPS at mAP 0.8668; its 2.48 ms (3.3%) overhead over uncalibrated DPM-Solver++ buys the highest accuracy in the comparison. The distilled three-stage model reaches 44.10 ms / 22.7 FPS at mAP 0.8595. Thus its $2.73\times$ acceleration is measured against the 120.34-ms RF+Heun reference, whereas its 0.0003 accuracy loss is measured against the 0.8598 six-stage DPM-Solver++ teacher. The

TABLE IX

SAME-HARDWARE MODEL-FORWARD LATENCY BENCHMARK ON DATASET 2 AT 512×512 RESOLUTION (RTX A6000, BATCH SIZE 1, 10 WARM-UP AND 500 TIMED CUDA-EVENT ITERATIONS). ACCURACY FOLLOWS THE PRESPECIFIED FINAL PROTOCOL; ONLY LATENCY IS HARDWARE-DEPENDENT.

Model	Solver	NFE	CHE	Latency (ms)	FPS	mAP
RF+Heun	Heun	7	42	120.34	8.3	0.8566
DPM-Solver++	DPM++	4	24	75.20	13.3	0.8598
KaryoFlow+LQCR	DPM++	4	24	77.68	12.9	0.8668
Distilled KaryoFlow (3-stage)	DPM++	4	12	44.10	22.7	0.8595
DPM-Solver++ +Top-K ($K=200$)	DPM++	4	24	70.57	14.2	0.8590
DPM-Solver++ +Top- K ($K=100$)	DPM++	4	24	68.90	14.5	0.8480
DINO R50	—	1	—	34.33	29.1	0.8649
RTMDet-L	—	1	—	33.83	29.6	0.8609
Cascade R-CNN	—	1	—	20.76	48.2	0.8527
DiffusionDet	Euler	1	—	24.09	41.5	0.8033
YOLOX-S	—	1	—	10.04	99.7	0.7951

cascade head dominates 90%+ of latency; the backbone+neck is a minor cost (~ 5.6 ms). These speed-linked re-evaluations differ by at most 0.0030 from the corresponding estimates in Tables III and V, consistent with stochastic proposal initialization. The same trained model supplies both accuracy and latency for the RF+Heun operating point.

The efficiency analysis separates three counts that are often conflated: DPM-Solver++ reduces solver NFE ($7 \rightarrow 4$), and head distillation reduces CHE ($24 \rightarrow 12$ at four NFE). Renewal removal does not change either count; it makes multistep history coherent and eliminates proposal replacement overhead.

V. DISCUSSION

KaryoFlow's components address different consequences of metaphase-image structure. The low-dimensional linear path and direct box prediction avoid explicit terminal error amplification and motivate a stable box-space parameterization, although no separate accuracy gain is claimed. Proposal identity matters because dozens of adjacent candidates evolve simultaneously, while the nearly fixed chromosome count permits a redundant proposal pool that makes renewal unnecessary at the retained operating point. LQCR then addresses

the remaining decision bottleneck: class confidence is already sufficient at IoU 0.5, but it does not identify which lower-area or overlapping boxes will survive stricter localization thresholds. The concentration of LQCR gains at AP90 and AP95 is consistent with this task-derived mechanism.

The efficiency results reveal complementary deployment controls rather than a single opaque speedup. DPM-Solver++ reduces solver evaluations, distillation reduces cascade-head evaluations, and Top- K reduces proposals after the first step. This decomposition enables an accuracy-oriented setting (0.8668 mAP, 77.68 ms) or a compressed setting (0.8595 mAP, 44.10 ms) without changing the central trajectory-ranking formulation.

The study has three boundaries. First, the method addresses chromosome detection within automated karyotyping; segmentation, straightening, pairing, and abnormality diagnosis are outside its present scope. Both cohorts are public chromosome datasets and lack patient identifiers, so patient-level leakage and clinical workflow outcomes cannot be assessed. Second, LQCR is evaluated through three independently trained parent-child pairs on each dataset, but three runs provide descriptive variability rather than a high-powered significance test. Natural-image validation is required to establish broader generality. Third, history-consistent inference relies on proposal redundancy: the observed $K = 100$ failure should be treated as an operating constraint, not hidden by the high- K result. These limits do not affect the box-path derivations, but they define the present empirical scope.

VI. CONCLUSION

KaryoFlow aligns trajectory learning, multistep state identity, and final localization-aware ranking for dense chromosome detection. RF supplies the few-step generative foundation; direct data prediction provides a well-conditioned terminal objective; and the proposal-identity analysis makes DPM++ history mathematically coherent at the retained proposal count. The decision-stage contribution, LQCR, improves paired test mAP by 0.0067 across three Dataset 1 runs and by 0.0065 across three Dataset 2 runs. In the fixed-detector mechanism analysis, Dataset 2 AP90/AP95 improve by 0.0278/0.0300 without changing candidate boxes before ranking. Together with orthogonal inference compression, these results establish a competitive and interpretable generative detector for chromosome karyotyping.

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